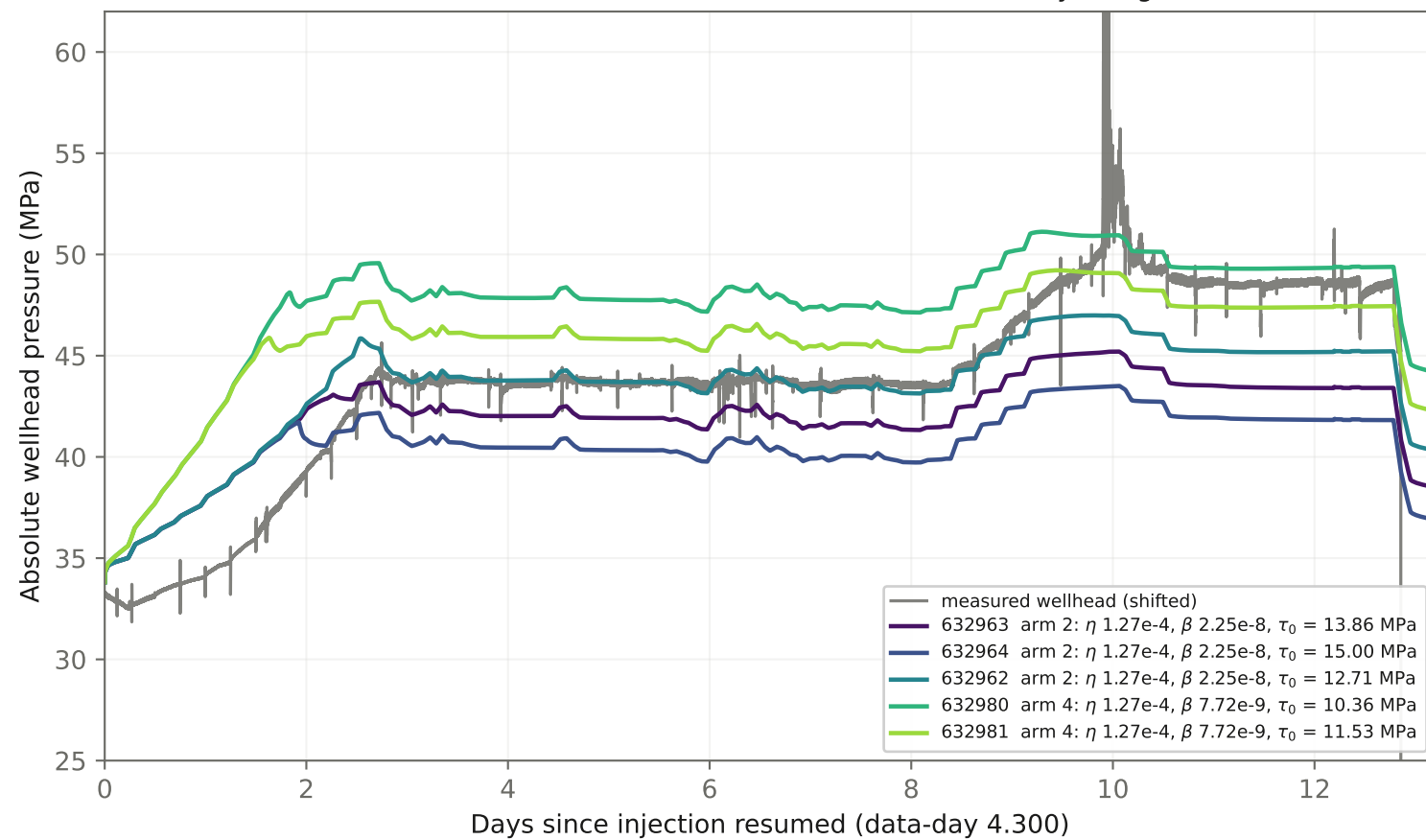
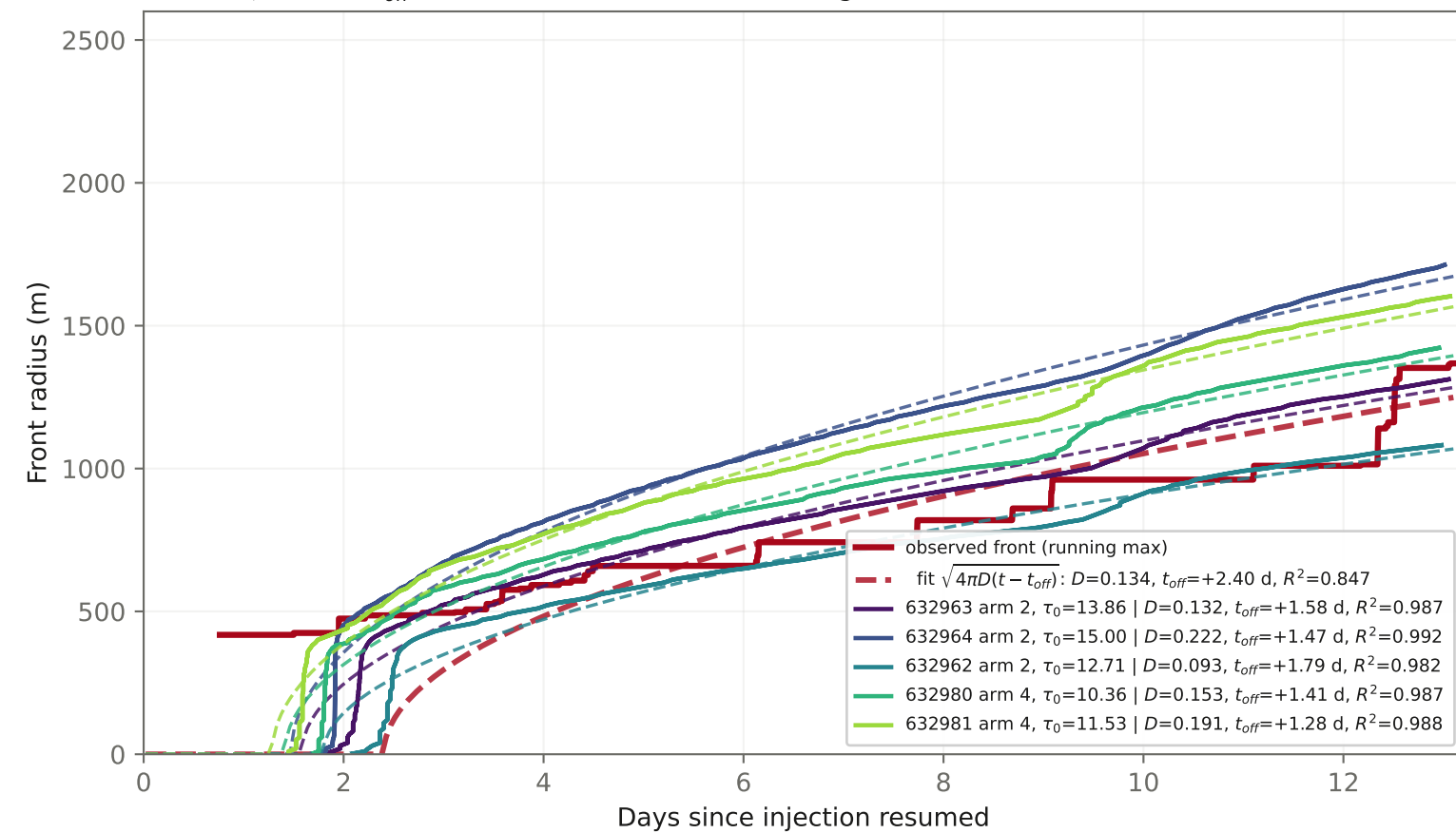


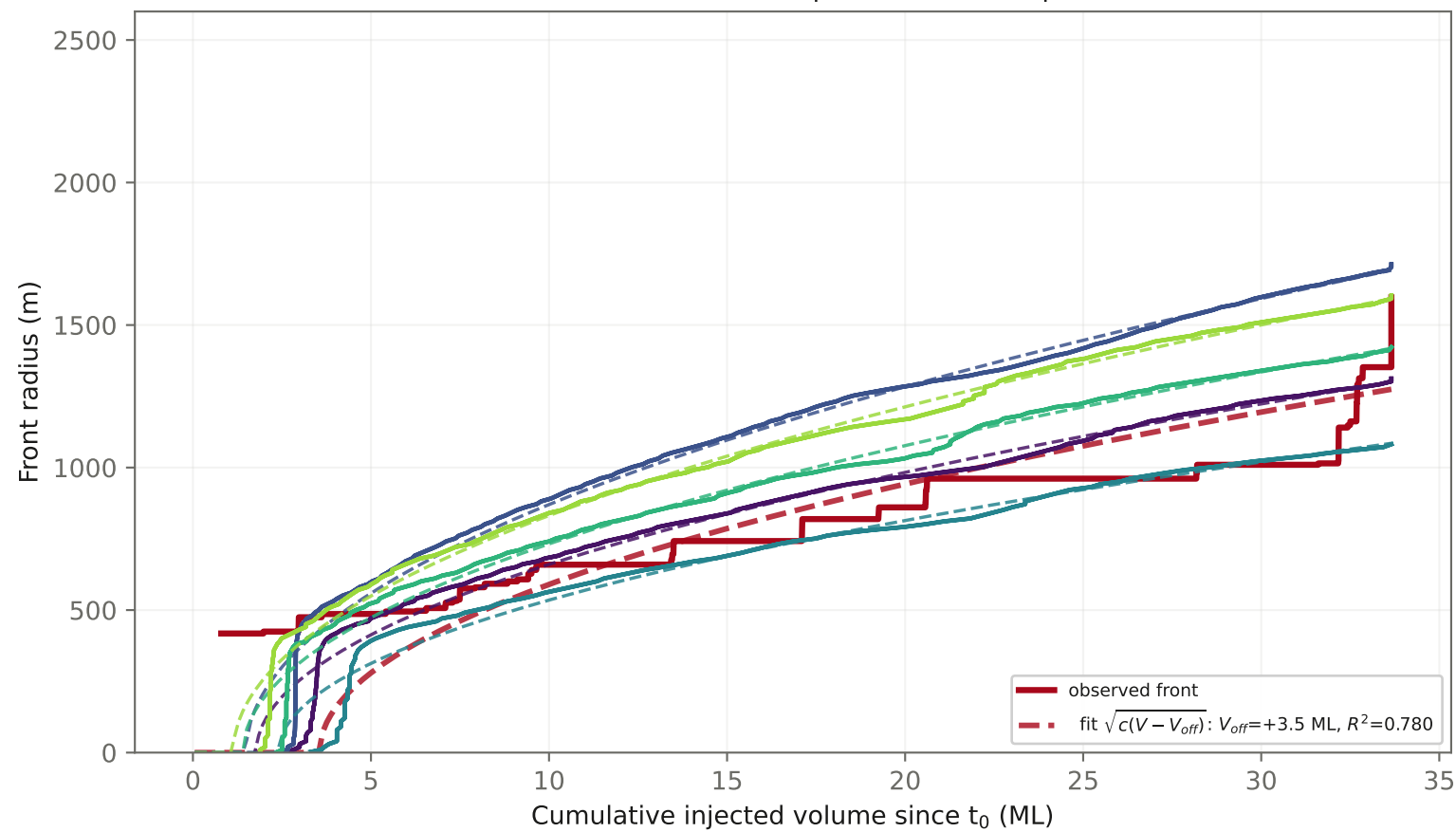
Wellhead pressure. Observed pulled back 4.300 d;  
the 87.8 MPa water-hammer transient is off-scale by design



R-T with square-root fits (dashed).  
 $r = \sqrt{4\pi D(t - t_{off})}$  — the  $\sqrt{t}$  law with a shifted origin, since neither front starts at  $r = 0, t = 0$



R-V with square-root fits (dashed).  
Volume removes the rate-step staircase that spoils R-T



Slip vs the observed INCREMENT since  $t_0$ .  
HBI zeroes slip at restart, so absolute slip is not the target

